

PRIYANKA RAJAN

Bengaluru, India

[Linkedin](#) | [G-scholar](#) | priyanka.rajan05@gmail.com | [GitHub](#)

Education

University at Buffalo-Roswell Park Comprehensive Cancer Center | Buffalo, NY, USA

Ph.D. Cancer Sciences 2019 - 2025

- ★ Thesis focus: Immuno-oncology and targeted therapy

M.S. Cancer Sciences 2017 - 2019

- ★ Thesis focus: DNA damage response to small molecule anti-cancer agents

PSG College of Technology | Coimbatore, India

BTech. Biotechnology 2013 - 2017

- ★ Thesis focus: Molecular mechanisms of cell division

Technical skills

Protein & Cell biology: Multiparametric flow cytometry (7-12 color panels), PBMC isolation, primary cell isolation, culture, sorting, western blot, Immunofluorescence, Immunohistochemistry, proliferation assay, migration, cell death, reporter assays

Molecular Engineering: CRISPR-Cas9, RT-qPCR, cloning, plasmid vector construction, PCR, restriction digestion, ligation, transformation, transfection, primer design, mini and maxi preps, sequencing

Computational & ML: FlowJo, FCSEXPRESS, MS Office, R (Shiny, TCGAbiolinks, biomaRt), Python (Jupyter), Linux/Bash, PK/PD modeling, high-throughput data analysis

In vivo Models: Syngeneic and PDX mouse models, drug efficacy testing

Experience

Research Associate (Machine Learning and bioinformatics)

Icahn School of Medicine at Mount Sinai, New York, NY (Remote) Apr 2025 - present

- ★ Developing unsupervised machine learning workflows to identify cell of origin for plasma disease biomarkers in high-dimensional transcriptomic datasets.

Graduate Research Assistant, Roswell Park Graduate Division Aug 2019 – Feb 2025

- ★ Manage daily laboratory activities including the maintenance of data, financial records & placing orders; plan, perform and analyze experiments; review scientific literature; propose & test treatment strategies with medical & research professionals
- ★ Oversee the design, implementation, and execution of all pre-clinical mouse model & tissue-cell culture experiments.

Senior Research Aide (Pharmacodynamics simulation and modeling)

Enhanced Pharmacodynamics LLC, Buffalo, NY Jan 2019 - May 2019

- ★ Developed an interactive web app using R Shiny, to predict immune toxicity of an anti-cancer drug candidate, based on existing population pharmacokinetic/pharmacodynamic (PK/PD) models.

Presentations

Translational Research Cancer Centers Consortium, Feb 19-21, 2025, Seven Springs, PA, USA.

Rajan, P., Zollo, R., Guo, Y., Lieberman, M., Alruwaili, M., Zonneville, J., Morreale, B., Olejniczak, S., Barbi, J., Abrams, S., Bakin, A. Targeting p38 to reverse immunosuppressive tumor microenvironment in metastatic breast cancer. (Oral & Poster)

American Association of Cancer Research, Apr 5-10, 2024, San Diego, CA, USA.

Rajan, P., Zollo, R., Guo, Y., Lieberman, M., Alruwaili, M., Zonneville, J., Morreale, B., Olejniczak, S., Barbi, J., Abrams, S., Bakin, A. Abstract 5536: The role of p38 MAPK in the tumor-induced immunosuppressive microenvironment in metastatic breast cancer. Cancer Res 84(6_Supplement), 5536-5536. (Poster)

Publications

[Rajan, P., Zollo R., Guo Y., Alruwaili M., Zonneville, J., Lieberman M., Morreale B., Olejniczak S., Barbi J., Abrams, S., & Bakin, A. *p38 blockade reverses the immune suppressive tumor microenvironment in metastatic breast cancer*. 2025. Under review at NPJ Precision Onc.](#)

[Alruwaili, M., Zonneville, J., Naranjo M., Serio, H., Melendy, T., Straubinger, R., Gillard, B., Foster, B.A., **Rajan, P.**, Attwood K., Chatley, S., Iyer, R., Fountzilas, C., Bakin, A.V. *A Synergistic Two-Drug Therapy Specifically Targets a DNA Repair Dysregulation That Occurs in p53-Deficient Colorectal and Pancreatic Cancers*. Cell Reports Medicine, 2024. 5\(3\): p. 101434.](#)

[Luzhin A., **Rajan P.**, Safina A., Leonova K., Stablewski A., Wang J., Robinson D., Isaeva N., Kanditze O., Gurova K. \(2023\). *Comparison of cell response to chromatin and DNA damage*. Nucleic Acids Research, 2023; gkad865.](#)

[Alruwaili, M., Guo, Y., Zonneville, J., Melendy, T., Straubinger, R., Foster, B.A., **Rajan, P.**, Withers H., Chatley, S., Iyer, R., Fountzilas, C., Bakin, A.V. *A Sequential Triple-Drug Strategy for Selective Targeting of p53-Mutant Cancers*. Under review, 2026.](#)

[Bharti A.C., **Rajan P.**, Jadli M., Pande D., Singh T., Bhat A. \(2018\). *Berberine as an Adjuvant and Sensitizer to Current Chemotherapy*. In Bharti A.C. & Aggarwal B. B. \(Eds.\), *Cancer Sensitizing Agents for Chemotherapy- Role of nutraceuticals in cancer chemosensitization* \(Vol. 2, 221-240, 2017\), Elsevier.](#)

Awards & Honors

- ★ Awarded for Academic Excellence by University at Buffalo
- ★ Supported by \$3000 grant award from University at Buffalo Mark Diamond Research Fund (UB-MDRF)
- ★ Selected as Career Experience Fellow by SUNY Research Foundation (SUNY-RF)
- ★ Selected for short-term project at Tata Institute of Fundamental Research (TIFR), Mumbai, India